Experiment 27

Aim

To perform cropping, copying, and pasting an image inside another image using OpenCV.

Procedure

1. Import the OpenCV library.
2. Read the source image and destination image.
3. Select a specific region from the source image using array slicing.
4. Crop the selected region from the source image.
5. Determine the position in the destination image where the cropped region has to be pasted.
6. Copy the cropped image into the selected region of the destination image.
7. Save the resulting image.
8. Display the original, cropped, and final images.

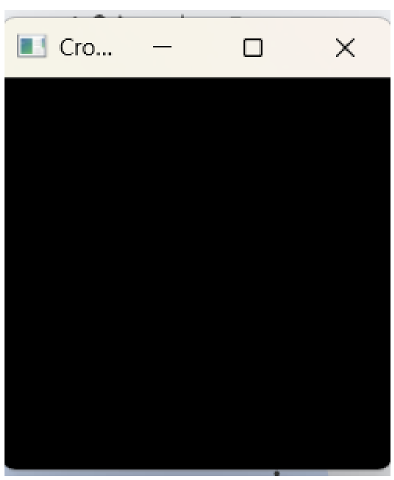
Code

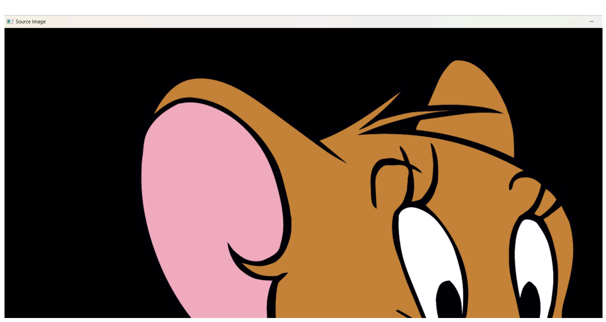
```
import cv2
source = cv2.imread("source.jpg")
destination = cv2.imread("destination.jpg")
if source is None or destination is None:
   print("Error: Image not found!")
else:
   crop = source[100:300, 100:300]
   h, w = crop.shape[:2]
   x, y = 50, 50
   destination[y:y+h, x:x+w] = crop
   cv2.imwrite("cropped_pasted_image.jpg", destination)
   cv2.imshow("Source Image", source)
   cv2.imshow("Cropped Image", crop)
```

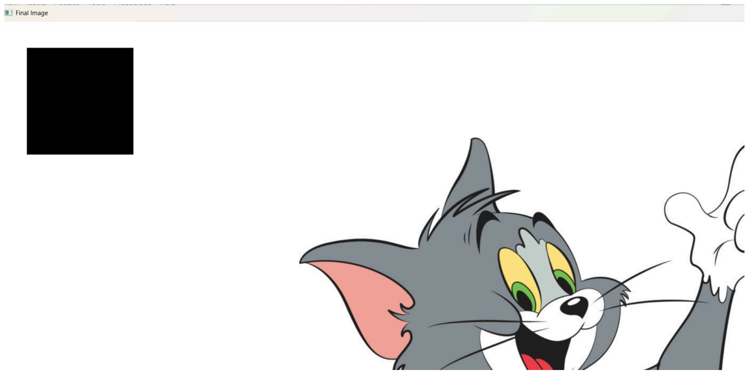
```
cv2.imshow("Final Image", destination)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:

Output:







Result

The watermark "MY WATERMARK" was successfully inserted into the image using OpenCV